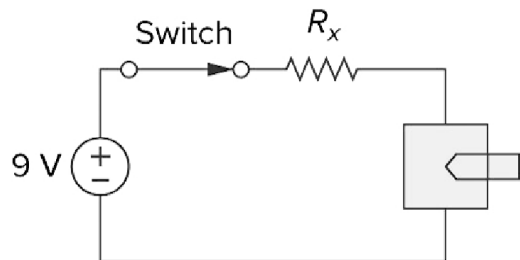


Q1 (2019 T3 Midterm Exam Q1)

An electric pencil sharpener rated 240 mW, 6 V is connected to a 9 V battery as shown in the figure below.

Calculate the value of the series-dropping resistor, R_x , needed to power the sharpener.

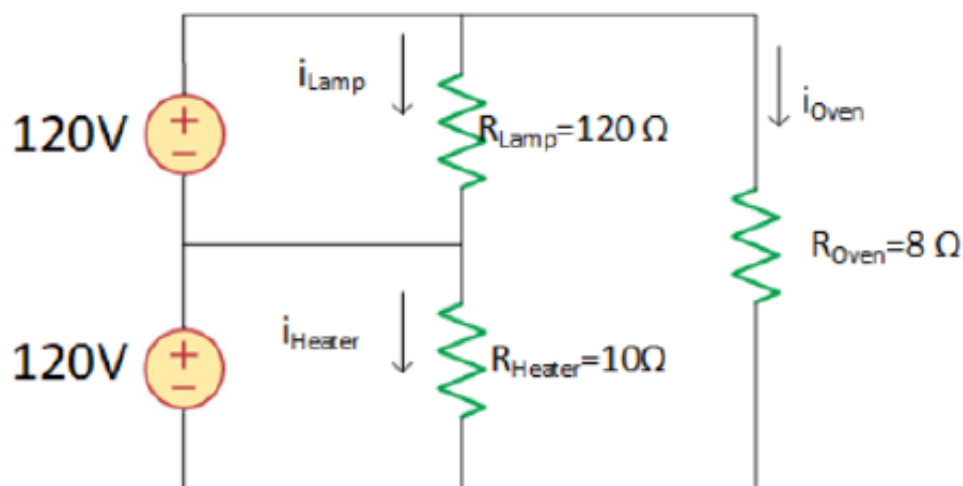


Answer: 75 Ω

Q2 (2022 T1 Midterm Exam Q1a)

The circuit in the figure below represents the DC equivalent of a simple residential power distribution system which allows the operation of appliances operating at 120V or 240V. Appliances are represented as resistors.

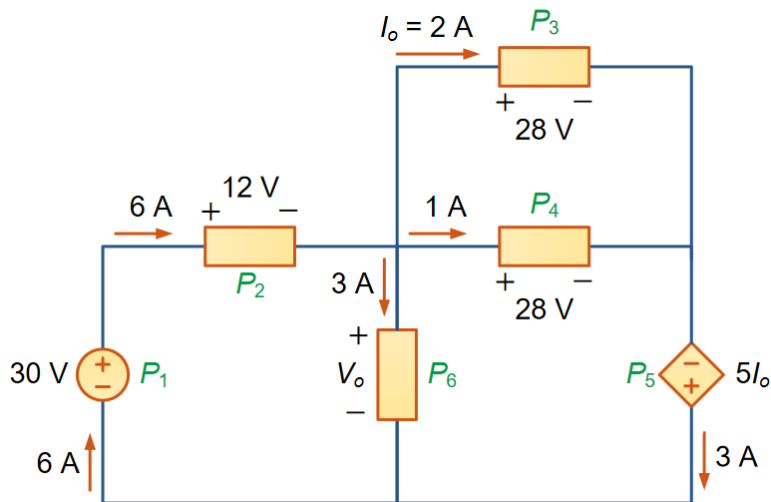
Calculate the current through each of the appliances.



Answers: $i_{lamp} = 1$ A, $i_{heater} = 12$ A, $i_{oven} = 30$ A

Q3 (Solved Examples-Topic 1 Q8)

In the circuit below, find V_o and the power for each element.



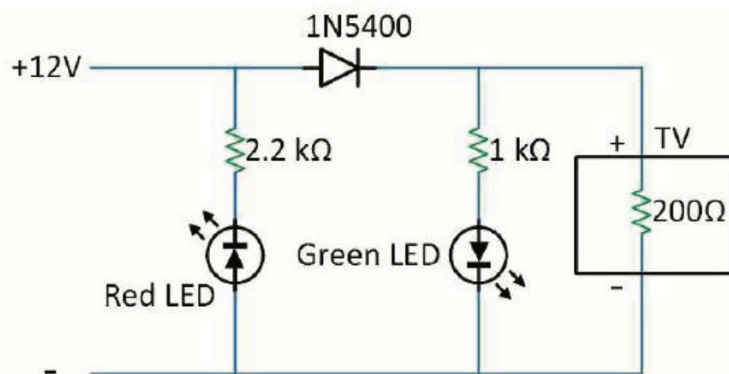
Answers: $V_o = 18\text{ V}$, -180 W , 72 W , 56 W , 28 W , -30 W , 54 W

Q4 (2021 T1 Final Exam Q1a)

The circuit below is a diode-based protection circuit for a portable TV. The red LED will glow to indicate “FAULT – battery reversed” in case of wrong connection to a DC power supply (reversed polarity). With the correct polarity, the green LED will glow to indicate “OK – normal operation”.

The voltage drop across the 1N5400 diode and LEDs under forward bias is 2.1 V . The TV is represented as a $200\ \Omega$ resistance.

- Calculate the current through the TV and green LED when the circuit is in normal operation.
- Calculate the current through the red LED when the circuit is in FAULT mode.
- Describe how the 1N5400 diode can protect the TV when the polarity is reversed.

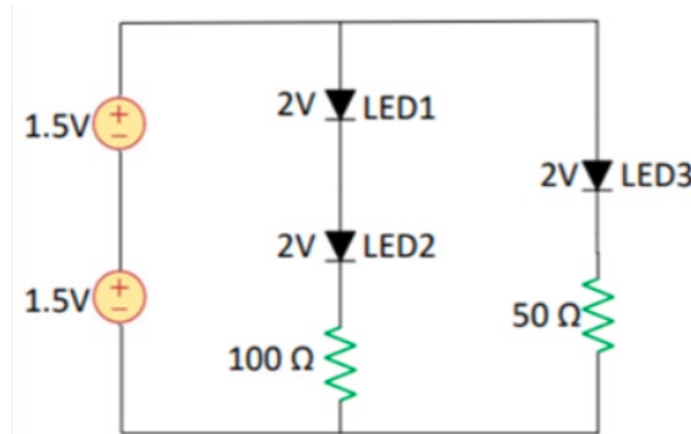


Answers: i) $i_{TV} = 49.5\text{ mA}$, $i_{green} = 7.8\text{ mA}$, ii) 4.5 mA

Q5 (2021 T3 Midterm Exam Q1b)

The circuit below is used to test an LED torch design. The torch is powered by AA batteries (1.5 V each) and has white LEDs with forward voltages across their terminals equal to 2 V. The maximum current that can flow through the LEDs is 25 mA. For currents higher than that, the LEDs get damaged and do not light up.

- Which LED(s) will light up in the torch? Please indicate the LED(s) name(s) clearly and support your answer with the appropriate calculations.
- Find the power dissipated in the 50 Ω resistor.



Answers: i) LED 3 is ON, ii) 0.02 W